

SOKKANAEM: Change-Gated State-Space Modeling with Cached Readout for Streaming Video Depth

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Abstract

Streaming video depth requires dense predictions on every frame, even when much of the image changes little. We present SOKKANAEM, a 4.19-million-parameter recurrent model that combines patch-change gating, selective state-space updates, temporal and spatial output caches, and a dense depth decoder. Binary gating preserves inactive temporal states, while periodic refresh and an all-active fallback limit reuse duration and handle high activity. We evaluate the frozen model with same-weight controls that separate state carry, retained-state readout, spatial context and output reuse. On four reserved TUM sequences, the reference model obtains clip scale-shift AbsRel 0.1869 at 256 frames, compared with 0.1837 for dense carry and 0.1849 for MiDaS Small. Its motion-referenced TCE is 0.0352 versus 0.0640 for MiDaS, although its scale variation is not uniformly smaller. The measured development PNG-to-depth latency is 7.496 ms/frame versus 7.521 for dense carry. Thus, computational reuse does not yet establish substantial pipeline acceleration. Conditional error and cache-age analysis explains the design trade-offs. The contribution is a compact streaming architecture and a controlled characterization of its accuracy, temporal behavior and implementation cost, rather than a claim of universal performance superiority.

Keywords: state-space model; conditional computation; video depth; error propagation; cache refresh; reproducible evaluation

1. Introduction

Video depth estimation supplies a dense geometric signal for each arriving RGB frame. Processing every frame independently discards temporal information and repeatedly evaluates similar regions. A streaming model should instead reuse useful history while maintaining depth quality, responding to scene changes and controlling per-frame execution cost.

We present SOKKANAEM, a compact recurrent video-depth architecture built around patch-level visual change. An external, untrained detector selects temporal state updates; separate temporal and spatial caches reuse block outputs; a dense decoder still predicts depth at every pixel. The input is the current frame and a stream-specific state dictionary, so the native prediction path does not require future frames, optical flow or ground truth. Fixed-period refresh and a high-activity fallback complement the change gate.

The central design question is how to combine temporal memory and computational reuse without confusing a held internal state with an unchanged or accurate output. Our

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model exposes state carry, readout reuse, spatial caching and refresh as distinct mechanisms. This makes their effects testable with the same trained weights rather than attributing every difference to a single sparsity score.

Our contributions are (i) a 4.19M-parameter streaming architecture combining externally gated selective SSMs with cached readout and dense depth prediction; (ii) same-weight, matched-mask interventions identifying the roles of retained-state readout and spatial context; (iii) evaluation of aligned accuracy, motion-referenced temporal error, scale variation and input-to-output cost; and (iv) conditional design analysis of state error, cache age and refresh overhead. The existing perturbation inequalities support the architecture’s design rationale; they are not presented as new universal stability laws.

The frozen reference does not dominate all controls or baselines. In particular, its measured desktop pipeline latency is close to dense carry, and lower short-step variation does not imply lower long-clip scale variation. We distinguish these observed trade-offs from the design objective of faster, temporally reliable streaming inference.

2. Related Work

Skip RNN explicitly uses a binary update/copy operation [3]. Mamba introduces input-selective state-space dynamics and hardware-aware scanning [4]. We do not claim either binary preservation or input selection as a first contribution. DeltaCNN propagates sparse frame differences through CNN operators [5], and Eventful Transformers reuse computation based on changing tokens [6]. Change-based reuse itself is therefore also established. Spiking SSMs such as SPiKE-SSM [7] and SpikySpace [8] investigate sparse dynamics in sequence learning and forecasting. Their architectures and tasks differ from dense RGB video-depth readout, but they preclude a broad claim of the first sparse or event-driven SSM.

The distinction investigated here is the combination of externally selected patch updates, selective SSM states, separate output caches, and measured depth/pipeline error. The literature comparison is targeted, not an exhaustive priority search. We do not report cross-task numerical comparisons against those methods. DPT [9], MiDaS [10] and Depth Anything V2 [11] provide independent-image depth references, not a complete benchmark of causal and noncausal video-depth systems. Video Depth Anything (VDA) [1] adds a spatial-temporal depth head; online VDA [2] is a distinct online formulation. Historical VDA scores in this repository do not have frame-ID and weight/source-hash linkage sufficient for inclusion in the frozen comparison. The audited historical wrapper calls a 32-frame offline, unmasked temporal-attention path, not frame-causal inference. The supplementary audit preserves those results separately. A matched modern video-depth baseline remains a limitation, not a completed comparison.

3. SOKKANAEM: A Change-Gated Streaming Depth Model

3.1. Architecture and reference configuration

The model uses a stride-16 RGB embedding, four alternating temporal/spatial blocks, 192-dimensional tokens, inner dimension 384, state dimension 16, four-direction spatial scans, local spatial convolution, and a dense DPT-style binned depth decoder. It has 4,185,872 parameters. The baseline is the seed-0 EMA checkpoint at 256 pixels, with pixel-MSE hysteresis thresholds 0.05/0.025, keyframe interval 30, both output caches, and an all-active fallback above activity 0.4. GMC is disabled in the primary model. The recorded training ancestry contains 60k, 60k, 25k and 8k stage steps; this is not an independently verified optimizer-update count or evidence of teacher-free ancestry.

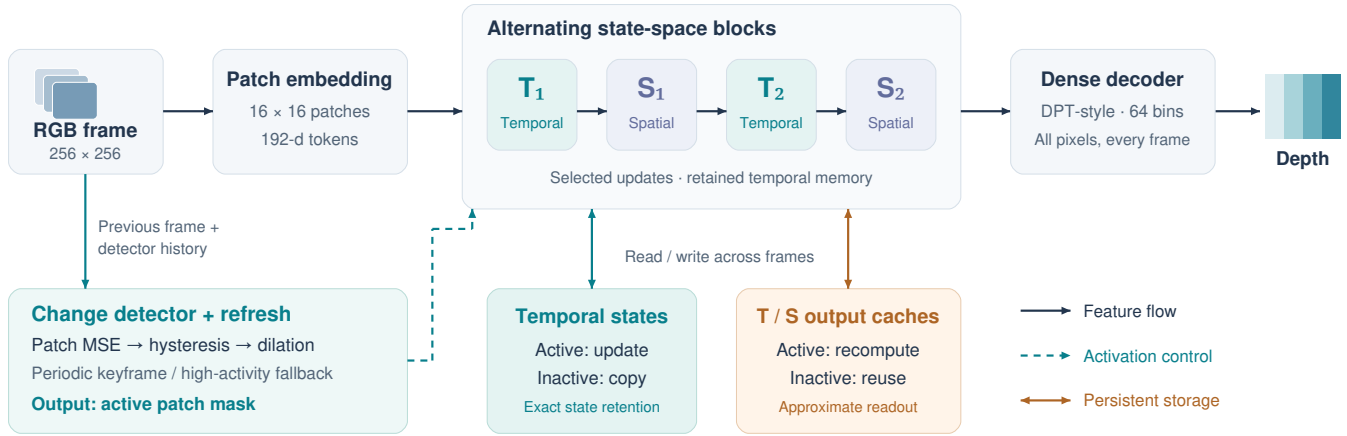


Figure 1. SOKKANAEM architecture. The upper path maps each RGB frame through patch embedding, four alternating temporal (T) and spatial (S) state-space blocks, and a dense depth decoder. The change detector supplies the activation mask (dashed arrows); bidirectional arrows denote access to persistent temporal states and T/S output caches across frames. Inactive temporal states are copied exactly, whereas cached readout and omitted spatial context introduce approximations. The depth icon is illustrative, not a measured prediction. Arrows show conceptual data/control flow rather than kernel execution order.

3.2. Change selection and refresh

For patch i with Cp^2 RGB values, the detector computes

$$d_{t,i} = \frac{1}{Cp^2} \sum_{u \in i} (I_t(u) - I_{t-1}(u))^2.$$

It uses threshold τ_{off} for a previously active detector patch and τ_{on} otherwise, followed by one-patch dilation. The first frame and frames whose zero-based index is divisible by K activate every patch. At batch one, the model additionally switches to all-active execution if the detector mask's mean activity exceeds 0.4. Detector hysteresis bookkeeping and the model's post-detector fallback are distinct; fallback does not rewrite the stored detector mask. These rules are fixed heuristics, not learned uncertainty estimates or an adaptive error certificate.

3.3. Temporal memory, cached readout and spatial context

Each temporal block maintains a hidden state for every patch. Without temporal output caching, an inactive hidden state is preserved but the block can still read it using current-input-dependent parameters. With temporal caching enabled, inactive block outputs are reused and the corresponding temporal computation is bypassed. These two modes separate state preservation from readout reuse.

Spatial blocks use four-direction scans and local spatial convolution. The spatial-cache path processes a selected token subsequence and restores dense block outputs by combining updates with cached entries. Omitting inactive spatial context can affect active tokens as well; this path is an approximation, not dense-network equivalence. The decoder fuses intermediate block features with RGB information and emits a full depth map at every step. Sparse backbone updates therefore do not make the embedding or decoder free.

3.4. Streaming execution and training

A stream starts with empty temporal states and caches. For each arriving frame, the model (1) selects the detector mask and applies any forced full update, (2) embeds the frame, (3) traverses temporal and spatial blocks with the selected state/cache policies, (4) decodes dense depth, and (5) returns the updated stream state. Separate state dictionaries

allow interleaved streams to share weights. Neither the predictor nor its change detector consumes the evaluation GT or RAFT flow.

We evaluate the existing v11 seed-0 EMA checkpoint, not a newly trained variant. Its recorded final stage uses 8000 steps, length-24 clips, batch two, random masks with maximum skip fraction 0.5 and the recorded depth/gradient/normal/bin/warp/edge/spread loss settings. The training mask is not the deployment change detector, leaving a train-deploy mismatch. Full settings and ancestry limitations are in the reproduction appendix. No adaptive-refresh policy or improved encoder/decoder is claimed to have been trained or evaluated in this revision.

4. Design Rationale and Conditional Guarantees

4.1. State preservation and approximation

For a shared token input, write $F_t(h) = \Phi_t h + q_t$, with $\Phi_t = \text{diag}(\exp(-\lambda_j \Delta_{t,j}))$ and $q_{t,j} = \Delta_{t,j}(B_t x_t)_j$. Binary gating implements $\tilde{h}_t = M_t F_t(\tilde{h}_{t-1}) + (I - M_t)\tilde{h}_{t-1}$. Zero step preserves state, but the input term is a first-order approximation to exact ZOH. Current-input-dependent readout and spatial-context omission need separate analysis.

With $e_t = \tilde{h}_t - h_t$ and $r_t = (I - M_t)((I - \Phi_t)\tilde{h}_{t-1} - q_t)$, the exact shared-input identity is $e_t = \Phi_t e_{t-1} + r_t$. Products of transition norms bound the propagated initial error and injected defects. A uniform geometric bound additionally requires a uniform contraction and a valid defect bound. Different deep-layer token trajectories add parameter/input perturbations; a negative diagonal A alone does not certify the full network.

The first temporal block admits a conditional cache bound $L_R \|W\| a \sqrt{C p^2 \tau_{\text{on}}}$ after a inactive steps, provided the state-fixed readout has the stipulated Lipschitz constant. It is not a tau-only depth guarantee. Keyframes bound cache age by $K - 1$ and remove the local approximation at the same incoming state, but they do not recover the state of a continuously dense history. Full assumptions, proofs, counterexamples and the no-fit AbsRel connection are in Appendix A.

4.2. Paid computation

Under the explicitly additive model $C_d = F + V$ and $\bar{C}_s = F + \bar{a}_{\text{eff}} V + \bar{H}$, normalized speedup is $[f + (1 - f)\bar{a}_{\text{eff}} + \omega]^{-1}$. Overhead must be smaller than saved variable cost for acceleration. This is Amdahl-style reasoning [12], not a conversion of MAC reductions into observed time. Keyframes and fallback events are counted as a union. At 256 frames and $K = 30$, even a static stream has nine keyframes. The analysis links two constraints on refresh: a certified cache-only tolerance would impose an upper bound on K , whereas a target speedup in the fixed-cost model imposes a lower bound. Their intersection is a conditional design check, not a selected operating point or an accuracy certificate. The constants needed for a useful numerical cache guarantee have not been certified for this network.

5. Experimental Setup

5.1. Development and reserved transfer sequences

The previously reused TUM/Bonn holdout is development data, not an independent test. It contains 488 L8 clips, of which 487 have scorable GT, and 13 L256 clips (3328 frames) from four sequences. The final evaluation uses four reserved TUM Freiburg3 sequences: sitting_xyz, sitting_rpy, walking_xyz and walking_rpy. The frozen lists contain 462 L8 clips and 13 L256 clips. These are new sequences in the same capture setting; they do not establish scene-disjoint or fixed-camera generalization. L32 is not evaluated in this final protocol. The lengths share underlying footage and are not independent samples. RGB/depth timestamps are paired within 20 ms, with possible reuse of a depth

frame. Sensor validity, resize/crop and depth scale follow the frozen evaluation contract and the TUM data documentation [13]; Bonn supplies dynamic indoor development sequences [14].

5.2. Comparisons and scoring

Final conditions were fixed before reserved predictions: K30, K5, dense carry, dense reset, output hold, MiDaS v2.1 Small at 256, DA2 Small at common target 256 (actual input 252), and DPT-Large at common 256. No model was retrained or promoted after test results. Native controls share the frozen weights; external baselines have different training data and budgets, whose overlap is not fully known. DPT’s four missing checkpoint parameters belong to an unused branch; execution hooks reject any use of that branch.

All models score the same 256-pixel ROI. Native metric outputs have a no-GT-fit panel; relative models are not relabeled as metric. Median and disparity scale-shift alignment are separate panels. The main shape comparison uses one fit per clip. Frame-fit L256 scores are supplementary shape diagnostics without temporal scores. Nonpositive fitted disparity is counted as an alignment failure; the resulting errors remain in aggregates. The frozen inverse-disparity conversion uses $1 / \max(d, 10^{-3})$, so an invalid fitted disparity can produce a 1000-unit fallback depth. This is not a physical depth estimate; it explains the sensitivity of RMSE to failed fits and is reported rather than removed. Nonfinite predictions fail execution. No-GT clips are explicitly counted, not substituted. Optical-flow-based temporal error uses a shared RAFT estimate and the frozen scoring equations; flow and GT are never fed to a predictor. Dynamic masks are a flow-residual heuristic, not semantic annotations.

5.3. Statistics and timing

Source pixel-pooled and equal-sequence means are reported separately. Paired sequence bootstrap enumerates 4^4 ordered resamples, while sign flipping enumerates 2^4 cases. The minimum possible two-sided sign-flip p is 0.125. Intervals are exploratory, unadjusted for multiple comparisons, and not finite-sample coverage guarantees. Frames, neighboring clips and repeated timing runs are not independent sequence replicates. The three existing training seeds differ only in the final 8k stage from a common parent.

Timing is development-only: the first L256 clip of each development sequence, 1024 matched frames, batch one, FP32 eager, TF32 off, RTX 4090, one complete warmup and five repeats. The boundary includes warm-cache PNG read/decode, crop/normalization, transfer, actual detector/fallback/keyframes, model, interpolation and CPU output copy. Model loading, GT, flow, fitting, scoring and output-file writes are excluded. A separate synchronized pass measures components. No final-test accuracy is paired with development latency as a same-quality speedup. No edge timing or energy claim is made.

6. Model Evaluation and Ablation Studies

6.1. Frozen final transfer evaluation

Table 1. Final TUM L256: clip scale-shift alignment, source pixel-pooled. AbsRel, RMSE and TCE lower is better; δ_1 higher is better. Fail is the number of clips with any invalid fitted disparity; failures remain in aggregates.

Model	AbsRel	RMSE	δ_1	TCE	Fail
Sparse K30	0.1869	1.3742	0.7382	0.0352	0/13
Sparse K5	0.1828	1.3531	0.7468	0.0357	0/13
Dense carry	0.1837	1.3488	0.7465	0.0328	0/13
Dense reset	0.1909	1.3778	0.7412	0.0348	0/13
Output hold	0.1955	1.3877	0.7332	0.0345	0/13
DA2 Small 252	0.2166	2.1519	0.6686	0.0709	2/13
MiDaS Small 256	0.1849	1.3476	0.7369	0.0640	0/13
DPT-Large 256	0.1894	1.3559	0.7201	0.0690	0/13

Table 2. Final TUM L8: clip scale-shift alignment, source pixel-pooled. AbsRel, RMSE and TCE lower is better; δ_1 higher is better. Fail is the number of clips with any invalid fitted disparity; failures remain in aggregates.

Model	AbsRel	RMSE	δ_1	TCE	Fail
Sparse K30	0.1840	10.1931	0.8017	0.0578	7/462
Sparse K5	0.1837	10.1759	0.8032	0.0590	6/462
Dense carry	0.1946	10.9170	0.8020	0.0572	7/462
Dense reset	0.1669	8.6731	0.7979	0.0503	8/462
Output hold	0.1677	8.5613	0.7965	0.0484	8/462
DA2 Small 252	0.2074	16.9349	0.8305	0.1285	45/462
MiDaS Small 256	0.1539	9.5211	0.8270	0.0716	22/462
DPT-Large 256	0.1810	10.4890	0.8402	0.1004	9/462

These panels measure aligned relative shape, not calibration-free metric depth. The selected K30 model is retained irrespective of its test ranking. All 13 L256 and 462 L8 clips were processed; 0 clips had no scorable GT.

Table 3. Final L256 native K30: alignment gauges must not be conflated.

Gauge	AbsRel	RMSE	δ_1	TCE
none	0.2014	1.3372	0.7553	0.0468
median	0.2094	1.3231	0.7416	0.0480
scaleshift	0.1869	1.3742	0.7382	0.0352

Table 4. Exploratory equal-sequence L256 AbsRel differences (K30 minus comparator), $n = 4$. Un-adjusted percentile sequence-bootstrap intervals; exact sign flips.

Comparator	Difference	95% interval	p
Dense carry	+0.0023	[-0.0052, +0.0094]	0.625
Sparse K5	+0.0035	[-0.0007, +0.0082]	0.375
MiDaS Small 256	+0.0054	[-0.0292, +0.0359]	0.625
DPT-Large 256	+0.0011	[-0.0398, +0.0430]	0.750

6.2. Development diagnostics and pipeline cost

Across 1024 development frames, all 32 post-initial keyframes have zero same-state local defect. Nevertheless, the median concatenated hidden/cache state RMS differences

at those keyframes are 0.932, 0.490, 0.721, 1.069 in manifest sequence order. These are internal-state units, not depth errors. Cache age never exceeds 29. The FP64 first-block shared-input shadow bound holds at all 1024 steps; this does not certify FP32 end-to-end error. Full sequence names and frame traces are in the supplement. On the four matched development clips, sparse K30 takes 7.496 ms/frame versus 7.521 for dense carry, a 0.33% reduction. MiDaS Small takes 9.439 ms with matched-subset AbsRel 0.1637 versus 0.1630 for K30. That operating-point comparison is not statistical equivalence: on all 13 development L256 clips, K30/MiDaS AbsRel is 0.2091/0.1749. The native sparse stream uses 13.501 MiB of persistent state versus 12.000 MiB dense. Read/decode/preprocessing accounts for 69.4% of a separate synchronized profile. The associated idealized fixed-input ceiling is 1.44 times that profile, not a measured speedup. Existing final-stage seed replicates give source-balanced scale-shift AbsRel 0.1146 ± 0.0007 on development L8 and 0.2040 ± 0.0077 on development L256 (mean and sample SD, three seeds sharing the earlier training stages).

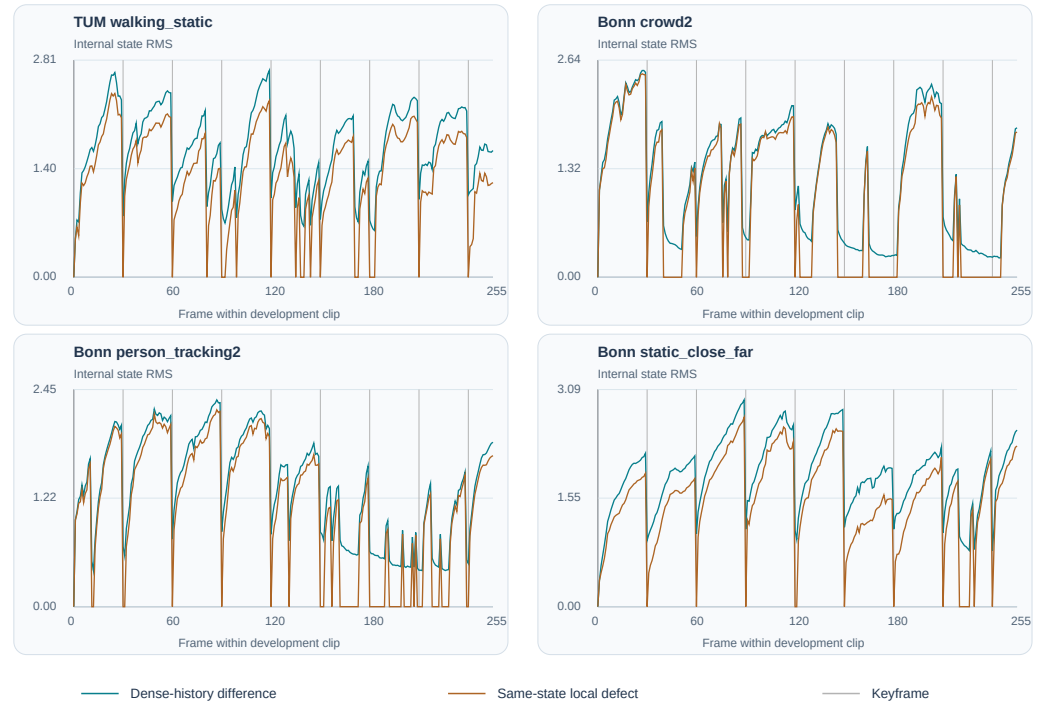


Figure 2. Development-only extended-state diagnostics. Dotted lines mark keyframes. The same-state local defect vanishes at refresh, while the difference from the separately evolved dense history persists. Curves are RMS over concatenated hidden states and caches, not GT depth error or uniquely allocated memory.

6.3. Model-component ablations

Table 5 restores the existing development ablations to the main model narrative. All seven conditions replay the reference K30 post-fallback mask on the same 13 L256 clips (3328 frames); mean activity is 24.18%. Scores are source pixel-pooled within TUM/Bonn, then source-balanced, using one disparity scale-shift fit per clip. Invalid fits remain included. These are inference interventions with one checkpoint, not separately optimized or retrained competing architectures.

Table 5. Development-only model ablations under a shared mask. Lower AbsRel/TCE is better. GMAC counts executed Linear/Conv2d operations only, excluding selective scan and other operations; it is not total FLOPs or latency.

Configuration	AbsRel	TCE	Failed clips	Partial GMAC
Both caches (reference replay)	0.2091	0.0435	2/13	0.656
Retained-state readout + spatial cache	0.2079	0.0441	2/13	0.804
Token-drop readout + spatial cache	0.2317	0.0538	3/13	0.804
Retained-state readout, no output caches	0.1863	0.0288	0/13	1.395
Token-drop readout, no spatial cache	0.3207	0.0511	0/13	1.395
Temporal cache only	0.1937	0.0275	0/13	1.247
Stateless predictor + output hold	0.1953	0.0312	0/13	0.990

With spatial caching enabled and temporal caching disabled, retained-state readout has lower AbsRel and TCE than zeroing its inactive temporal residual contribution. This supports the role of readout under that intervention, not universal superiority over token-dropping systems. Removing spatial caching improves these depth scores while increasing counted work. The simple output-hold control also improves these scores over reference replay. Consequently, the present dual-cache configuration is a particular quality-reuse trade-off, not a demonstrated optimum. We omit the older ablation latency here because its timing subset and boundary differ from the complete accuracy evaluation and from the main PNG-to-depth measurements.

6.4. Post-hoc scale-variation diagnostic

To address the earlier self-review, we summarize statistics already stored during final evaluation. This analysis was added after test access, not preregistered as a primary endpoint. It includes all eight frozen models and both lengths without tuning or new inference. For each valid frame, $s_t = \text{median}(G_t) / \text{median}(D_t)$ uses the same GT-valid pixels. We report the within-clip sample coefficient of variation $\text{sd}(s_t) / \bar{s}$, sample standard deviation of $\log s_t$, and mean absolute log difference between consecutive retained valid frames. The scorer skips frames without valid GT and returns zero if fewer than two remain; these conventions are unchanged. Native models use no-fit depth; relative models use their median-gauge inverse-disparity depth, not an affine-fit depth. Away from numerical clamp activation, a positive constant clip scaling cancels from these statistics; an affine disparity shift does not. The stored scorer clamps the predicted median at 10^{-6} , the scale factor at 10^{-12} and the CV denominator at 10^{-6} .

Table 6. Post-hoc scale statistics: clip means within each sequence, then four-sequence means. The CV interval enumerates sequence-bootstrap resamples; it is exploratory and unadjusted. Full intervals and gauge/failure counts are supplied as CSV.

Model	CV L8	CV L256	L256 CV interval	Log-SD	Log-step
Sparse K30	0.0312	0.2185	[0.0770, 0.3601]	0.1800	0.0180
Sparse K5	0.0318	0.2131	[0.0677, 0.3585]	0.1752	0.0183
Dense carry	0.0312	0.2101	[0.0642, 0.3560]	0.1723	0.0169
Dense reset	0.0355	0.2463	[0.0677, 0.4248]	0.1903	0.0235
Output hold	0.0338	0.2466	[0.0683, 0.4250]	0.1905	0.0218
DA2 Small 252	0.0744	0.2477	[0.1753, 0.3201]	0.2418	0.0696
MiDaS Small 256	0.0603	0.1892	[0.1104, 0.2680]	0.1760	0.0559
DPT-Large 256	0.0774	0.2554	[0.1613, 0.3496]	0.2219	0.0704

A smaller scale statistic alone does not establish better depth, and per-frame fitting can hide scale errors that a metric deployment still incurs. L8 and L256 use partly different footage and reset schedules, so their difference is not an isolated causal effect of history

length. Local shape changes can also affect the scale medians. These summaries are not an additive decomposition of AbsRel into scale and shape terms. Confidence intervals containing a null difference do not establish equivalence.

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247

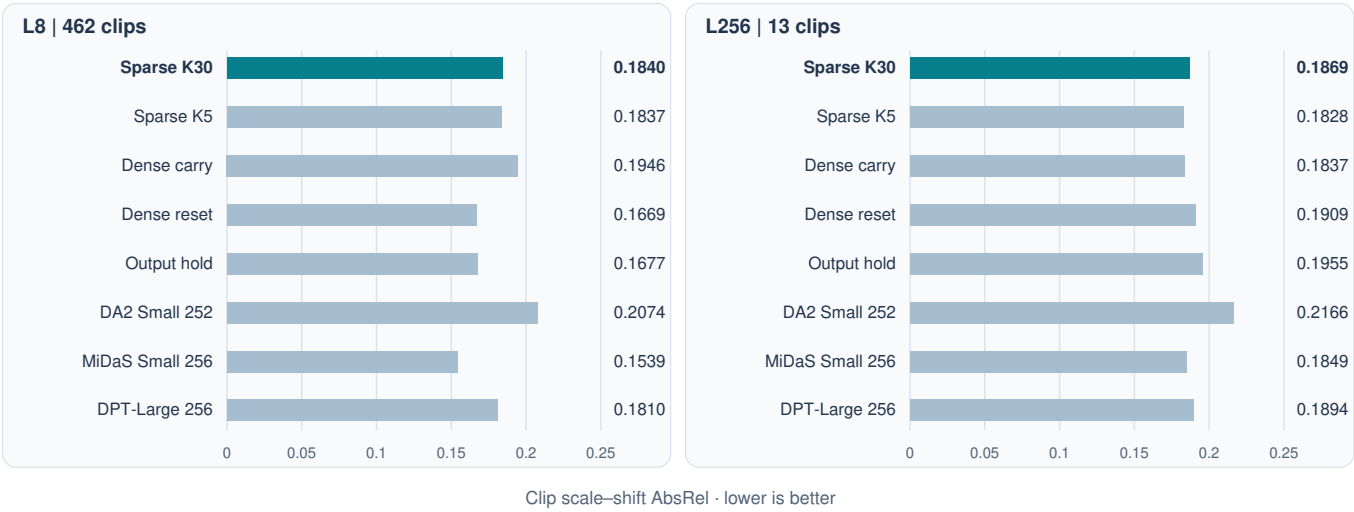


Figure 3. All eight frozen paths on L8 and L256, using the source pixel-pooled clip scale-shift AbsRel estimator of the main tables. Zero-based axes and all failed fits are retained. These point estimates have no sequence uncertainty bars: the exploratory paired sequence estimator is reported separately. The two lengths share some footage but are not identical history interventions.

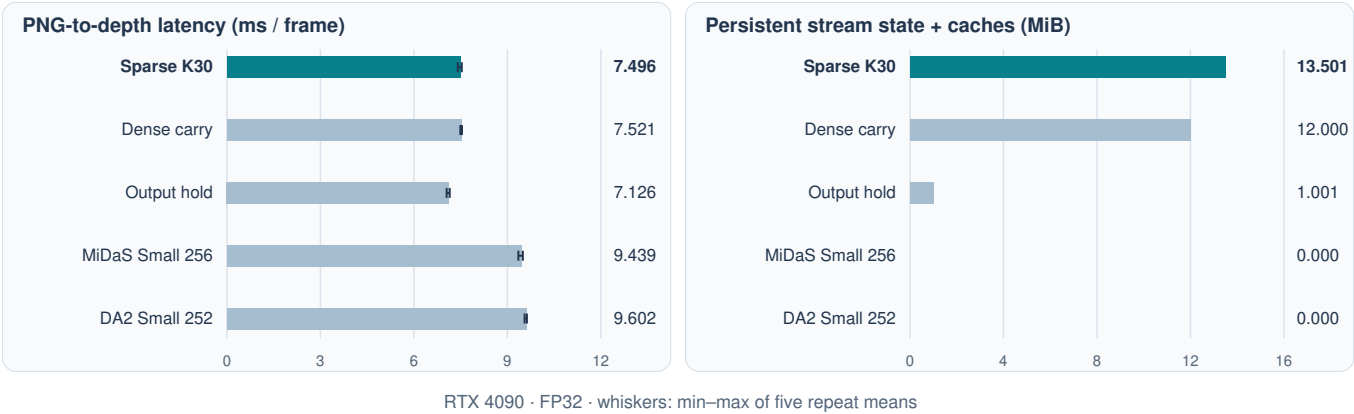


Figure 4. Development-only measured cost for five representative operating points. Latency includes the same PNG-to-depth boundary on four matched L256 clips; whiskers show the minimum/maximum of five repeat means, not confidence intervals. Persistent stream memory excludes weights and transient buffers; zero state in an image model does not mean zero total memory. Final-test accuracy is not used to claim matched-quality acceleration.

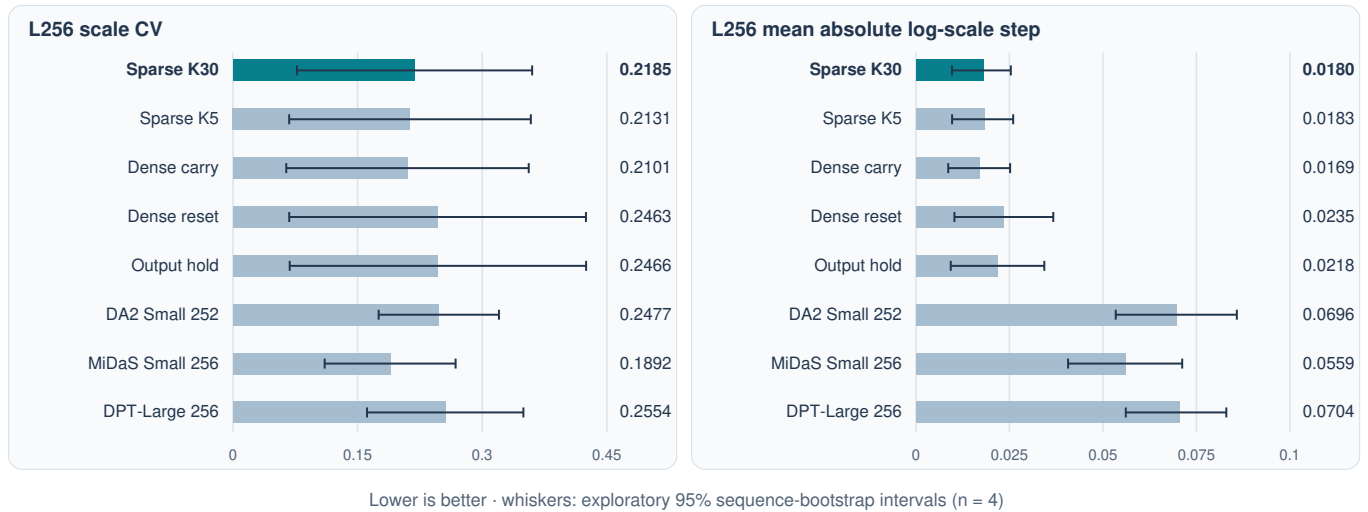


Figure 5. Post-hoc scale diagnostics for all eight frozen L256 paths. Four-sequence means and exploratory, unadjusted percentile sequence-bootstrap intervals; clip means are averaged within each sequence first. Native predictions use no-fit depth, while relative predictions use median-gauge depth. CV describes within-clip variation and log-step describes short-step variation; neither is a substitute for depth accuracy or proof of general temporal superiority.

6.5. Qualitative depth and temporal behavior

Figures 6 and 7 add post-hoc visualization of the already-opened evaluation, not a new independent test. Before generating these maps, we fixed the first L256 manifest clip of each of the four TUM sequences (IDs tum:0, tum:4, tum:7 and tum:10), spatial frame 127, and temporal frames 28–32 of tum:7. All frame indices are zero-based. We replayed SOKKANAEM, dense carry, MiDaS Small and output hold from each clip boundary; all 16 complete raw-prediction checksums exactly match the corresponding frozen results. No weights, thresholds or aggregate scores were changed.

For visual comparability, each model uses the existing single disparity scale-shift fit over the whole clip. This GT-assisted alignment can use later frames and is an evaluation-only operation, not a causal inference component or calibration-free metric output. All maps share the fixed 0–5 m display range; gray GT pixels are invalid, and values above 5 m saturate without being removed from the stored predictions or scores. No per-frame fitting or color rescaling is used. Full input paths, alignment coefficients, invalid-fit flags, framewise display clipping fractions and prediction hashes are supplied in `qualitative/provenance.json`.

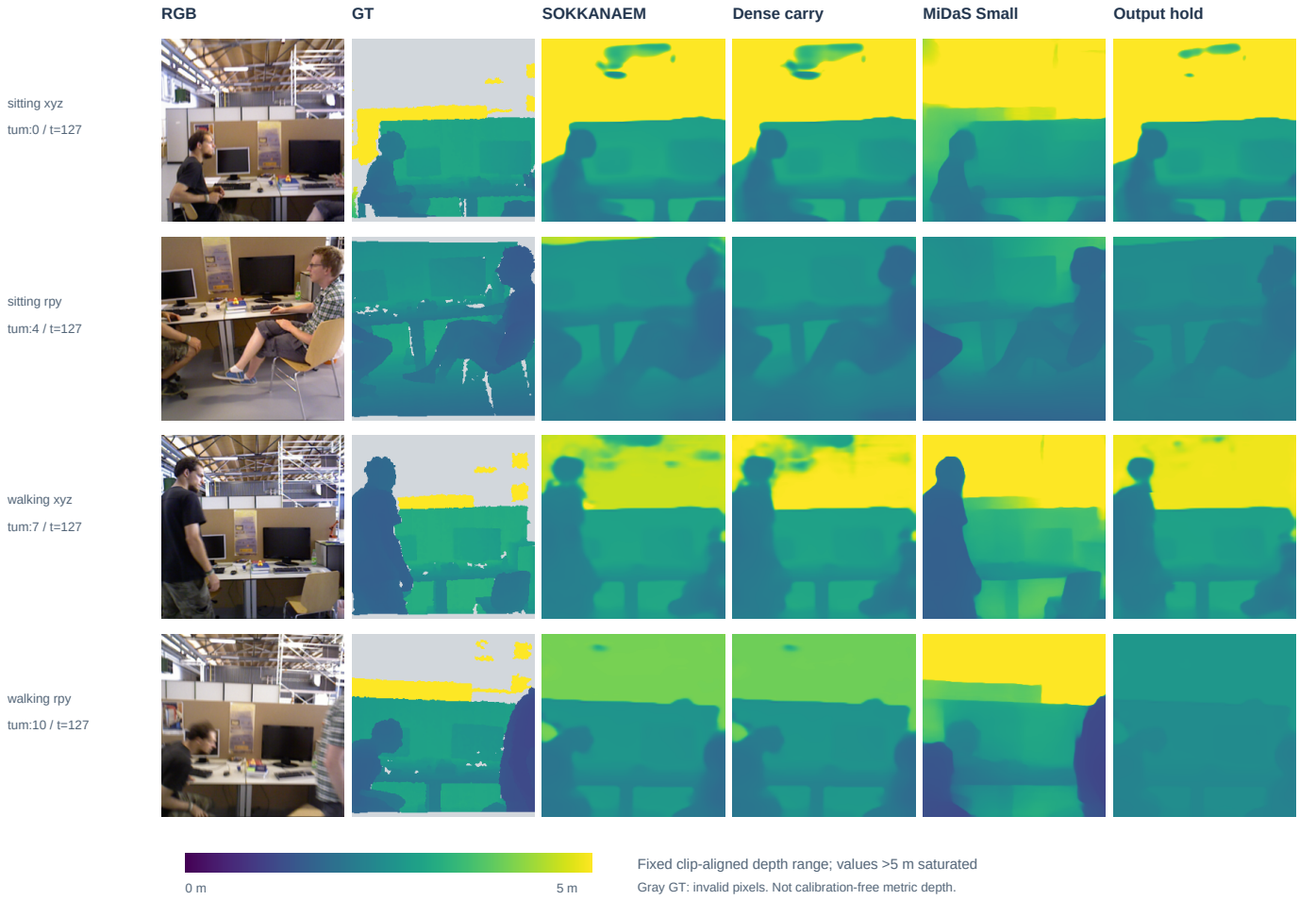


Figure 6. Post-hoc qualitative comparison at fixed frame 127 of the first L256 clip in each evaluated TUM sequence. Columns show RGB, valid GT, SOKKANAEM, dense carry, MiDaS Small and output hold. All models use one GT-assisted disparity scale-shift fit per clip and a common fixed 0–5 m color range. Gray GT regions are unobserved, not zero-depth targets; saturated backgrounds cannot establish agreement beyond the display range. The rows are fixed by manifest order rather than selected for favorable model appearance.

The spatial examples show similar large-scale layouts for SOKKANAEM and dense carry, but smooth outputs can suppress fine structures and person boundaries. MiDaS preserves a more distinct foreground silhouette in the walking-xyz example, whereas the native paths contain rounded artifacts in the distant background. On these displayed walking-xyz and walking-rpy frames, SOKKANAEM/MiDaS valid-GT AbsRel is respectively 0.2105/0.1196 and 0.2859/0.2028. The two sitting examples have the reverse ordering (0.0614/0.1309 and 0.0978/0.1087). These are illustrative single-frame values, not an additional aggregate benchmark. The selection includes tum:10, the highest original K30 clip AbsRel among the 13 final L256 clips, so the visual record does not exclude that adverse case. Predicted detail in invalid-GT regions cannot be validated from these images.

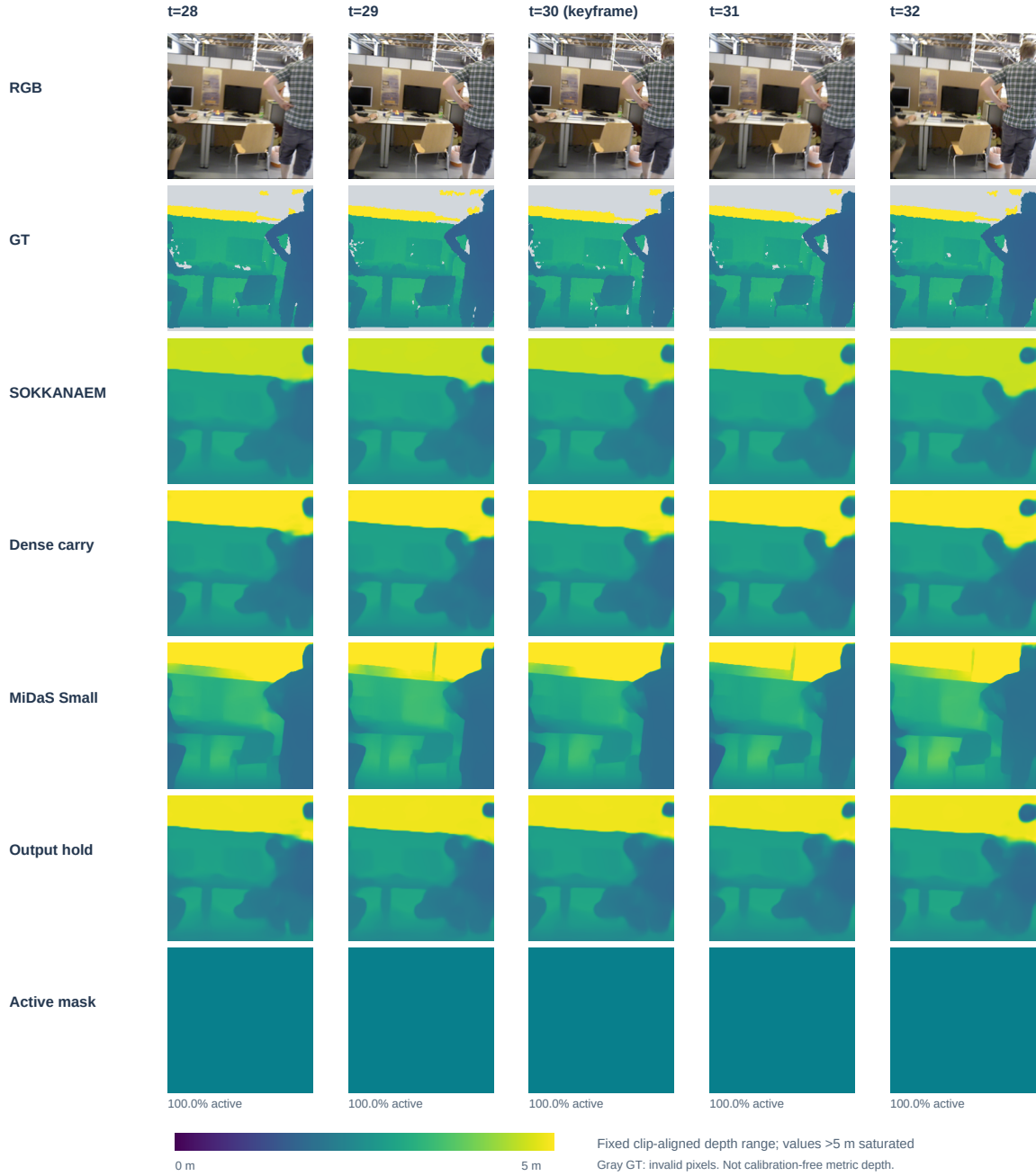


Figure 7. Five consecutive frames of the fixed walking-xyz clip tum:7 around scheduled keyframe 30, with the actual post-fallback SOKKANAEM activation mask (teal: active). All five masks are fully active, including the surrounding non-keyframes. Thus, this example documents high-activity execution and moving foreground boundaries; it does not isolate a sparse-reuse benefit or a causal keyframe effect. Depth alignment, invalid-GT handling and fixed display range are identical to Figure 6.

In the temporal window, the native paths remain visually similar while the moving person's outline is smoother than in the MiDaS maps. Because all five native masks are fully active, temporal similarity here must not be attributed to skipped updates. The supplement contains synchronized 256-frame videos for all four selected clips (qualitative/tum_0.mp4, tum_4.mp4, tum_7.mp4, tum_10.mp4). Playback is 10 frames/s for inspection, not measured inference throughput or the source capture rate. These visualizations complement, rather than replace, the quantitative temporal and accuracy measures. They do not establish general temporal superiority or performance beyond L256.

7. Discussion and Limitations

The central distinction is between preservation and approximation. A constant input can leave an ungated SSM approaching equilibrium while a skipped state remains frozen. Similarly, recomputing every cache entry on a keyframe does not erase the history carried by temporal states. Local defect elimination and global trajectory synchronization are different operations, as the development diagnostic demonstrates.

The shared-input bound is useful for identifying omitted innovations, but an observed small shadow-model residual does not certify the complete FP32 depth network. The pixel threshold does not control all deeper features, and the spatial subsequence path omits context even at active coordinates. Global Lipschitz constants have not been certified. Synthetic checks and this internal derivation audit do not replace mathematical review by the authors and an independent domain expert.

The external efficiency comparison also cannot isolate architecture from training. MiDaS Small and DA2 Small are larger than the native network, and we have not covered all similarly sized CNNs or modern causal video baselines. The present study supports bounded, implementation-specific statements rather than broad accuracy or efficiency superiority. More independent scenes and full-training replicates would be needed for stronger generalization and stability conclusions. Test access was a one-time finalization stage; any subsequent development would require a new untouched evaluation set. Our 1024 development diagnostic frames comprise four separate L256 clips, not one continuous L1024 trajectory. The older L1024 evidence concerns v9/v10, not the present v11; its PointOdyssey aggregate scores and frame curves use 20 and 8 clips, respectively. Nearly equal endpoints cannot establish stability throughout a trajectory. We therefore do not inherit the earlier continuous-stream or scale/shape decomposition claims. The added current-checkpoint qualitative replays cover four L256 clips; longer continuous evaluation remains outstanding.

8. Conclusions

SOKKANAEM combines patch-change selection, recurrent selective SSMs, cached block outputs and dense decoding in a compact streaming depth model. Controlled interventions show that retained temporal readout and spatial context influence depth quality independently of update activity. The evaluated model offers a specific accuracy–temporal-behavior operating point, but its current desktop pipeline does not establish substantial sparse acceleration and its long-clip scale variation remains a weakness. Conditional state/cache analysis explains why refresh and computational reuse require separate quality and cost checks. The next model-development priorities are reducing paid pipeline overhead and controlling stale-state/cache effects without sacrificing depth accuracy; their benefits remain to be established on an untouched evaluation set.

Supplementary Materials: The accompanying package contains complete per-model/per-sequence tables, frame-level development diagnostics, derivations, source hashes and reproduction commands. Raw third-party datasets and baseline weights are obtained from their original providers.

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Conflicts of Interest: AUTHOR TO COMPLETE: actual competing interests or a verified absence thereof.

Appendix A. Analytic Proofs

Appendix A.1. Implemented recurrence and zero step

Vectorize the inner-channel and state coordinates of one temporal SSM. For a shared exogenous token sequence, write

$$F_t(h) = \Phi_t h + q_t, \quad \Phi_t = \text{diag}(e^{-\lambda_j \Delta_{t,j}}), \quad q_{t,j} = \Delta_{t,j} (B_t x_t)_j, \quad (\text{A1})$$

where $\lambda_j > 0$ and $\Delta_{t,j} \geq 0$. A channel's step size is repeated over its state coordinates. The code uses $A = -\exp(A_{\log})$ and a softplus step size. Norms are Euclidean and induced operator norms unless specified.

Proposition A1 (Binary zero-step identity). *For a diagonal binary mask M_t , multiplying the step size by the mask gives*

$$\tilde{h}_t = M_t F_t(\tilde{h}_{t-1}) + (I - M_t) \tilde{h}_{t-1}. \quad (\text{A2})$$

Thus each inactive hidden-state coordinate is unchanged in real arithmetic.

Proof. On an active coordinate the original update is used. On an inactive coordinate the transition is $e^0 = 1$ and the input term is zero. These cases give the stated expression. Fractional masks do not generally have this update/copy interpretation because $e^{m\Delta A} \neq m e^{\Delta A} + 1 - m$. $\square$

The input term is first order, not exact zero-order hold (ZOH). With the input held constant, the ZOH coefficient is $(1 - e^{-\lambda\Delta})/\lambda$; see the ZOH definition in Mamba [4]. Its difference from the implemented coefficient obeys

$$\left| \Delta b - \frac{1 - e^{-\lambda\Delta}}{\lambda} b \right| \leq \frac{\lambda\Delta^2}{2} |b|. \quad (\text{A3})$$

Indeed, the coefficient difference is $\int_0^\Delta (1 - e^{-\lambda s}) ds \leq \int_0^\Delta \lambda s ds$. This local discretization error is distinct from skipping error: both paths below use the same implemented recurrence. State identity also does not imply readout identity, since the current input affects C, x, z, D and the residual. Finite CPU/CUDA tests supplement the real-arithmetic statement; they do not establish bitwise identity for exceptional values or all floating-point scan reassociations.

Appendix A.2. Skipping defects and state error

Theorem A1 (Shared-input defect propagation). *Let $h_t = F_t(h_{t-1})$ and let $\tilde{h}_t$ satisfy (A2), using the same Φ_t, q_t . Define*

$$e_t = \tilde{h}_t - h_t, \quad r_t = (I - M_t)((I - \Phi_t)\tilde{h}_{t-1} - q_t), \quad d_t = \|r_t\|. \quad (\text{A4})$$

Then $e_t = \Phi_t e_{t-1} + r_t$. If $\|\Phi_t\| \leq \rho_t$, then

$$\|e_T\| \leq \left(\prod_{j=1}^T \rho_j \right) \|e_0\| + \sum_{i=1}^T \left(\prod_{j=i+1}^T \rho_j \right) d_i. \quad (\text{A5})$$

Empty products equal one. In particular, if $\rho_t \leq \rho < 1$ and $d_t \leq \epsilon$, the bound is $\rho^T \|e_0\| + \epsilon(1 - \rho^T)/(1 - \rho)$.

Proof. Add and subtract $\Phi_t \tilde{h}_{t-1} + q_t$ in the gated update and subtract the dense update. Iteration gives a sum of propagated defects. Submultiplicativity and the triangle inequality give (A5); the uniform case follows by summing a geometric series. $\square$

A uniform lower bound $\lambda_j \geq \lambda_{\min} > 0$ and $\Delta_{t,j} \geq \Delta_{\min} > 0$ would yield $\rho = e^{-\lambda_{\min} \Delta_{\min}} < 1$ for the ungated comparison transition. The actual gated transition equals one on skipped coordinates. Without a useful uniform contraction bound, the non-expansive estimate $\|e_T\| \leq \|e_0\| + \sum_i d_i$ remains available.

Counterexample: zero input change.

For $F(h) = h/2 + 1$, equal zero initial states and all updates skipped, $\tilde{h}_t = 0$ while $h_t = 2(1 - 2^{-t})$, despite constant inputs. The error at $t = 4$ is 1.875. An all-active first frame does not remove the subsequent discrepancy. Thus a small pixel-change threshold alone does not bound the omitted innovation, even when the input change is exactly zero.

Different token trajectories.

If the gated branch uses $\tilde{\Phi}_t, \tilde{q}_t$, define its skip defect using those parameters. The error recurrence acquires the extra term $(\tilde{\Phi}_t - \Phi_t)\tilde{h}_{t-1} + \tilde{q}_t - q_t$. Its norm is at most $\|\tilde{\Phi}_t - \Phi_t\| \|\tilde{h}_{t-1}\| + \|\tilde{q}_t - q_t\|$. This term cannot be dropped for deeper blocks whose inputs depend on previously approximated features.

Theorem A2 (Whole-network conditional perturbation). *Fix the frame sequence and the masks selected by the sparse trajectory. Let z comprise all temporal states and block output caches, with the dense comparison overwriting an identically shaped set of caches at every step. Let F_t be the all-active network step and G_t the sparse step. Put $d_t^{\text{net}} = \|G_t(\tilde{z}_{t-1}) - F_t(\tilde{z}_{t-1})\|$. If F_t is L_t -Lipschitz on a region containing both states, then*

$$E_t^{\text{net}} \leq L_t E_{t-1}^{\text{net}} + d_t^{\text{net}}. \quad (\text{A6})$$

Equation (A5) holds with L_t, d_t^{net} in place of ρ_t, d_t . For dense and sparse output maps $P_t, \tilde{P}_t$, if P_t is $L_{P,t}$ -Lipschitz and $\eta_t = \|\tilde{P}_t(\tilde{z}_{t-1}) - P_t(\tilde{z}_{t-1})\|$, then $\|\tilde{y}_t - y_t\| \leq L_{P,t} E_{t-1}^{\text{net}} + \eta_t$.

Proof. Add and subtract $F_t(\tilde{z}_{t-1})$, or respectively $P_t(\tilde{z}_{t-1})$, and apply the assumed Lipschitz inequality. Iterate the state bound. $\square$

No whole-network $L_t < 1$ or practically small global constant is certified here. Negative diagonal A alone is insufficient: the elementary map $F(h) = 0.5h + u(h)$ with $u(h) = h$

expands by 1.5. This is a counterexample to an inference about feedback, not an identification of v11 with that map. Computing a counterfactual dense defect also costs work; the detector does not provide it as a free online certificate. Threshold-induced mask changes under perturbed images are outside this fixed-mask comparison.

For n fixed pixels with ground truth $d_i \geq d_* > 0$, the reverse triangle inequality gives the additional, no-fit statement

$$|\text{AbsRel}(\tilde{y}, d) - \text{AbsRel}(y, d)| \leq \frac{1}{n} \sum_i \frac{|\tilde{y}_i - y_i|}{d_i} \leq \frac{\|\tilde{y} - y\|_1}{nd_*}. \quad (\text{A7})$$

This does not remove the dense reference’s own ground-truth error. Separately refitted scale-shift alignment requires conditioning assumptions of its own; threshold accuracy metrics require margin assumptions. No new depth cutoff has been introduced into the evaluation protocol.

Appendix A.3. Cached readout and refresh

Proposition A2 (Conditional first-block cache bound). *Consider a pixel-gated patch with Cp^2 pixel values that remains inactive for $a = t - s$ steps after its last update. Suppose its embedding is $u = WI + b$ and the state-fixed block readout $R(u, h_s)$ is L_R -Lipschitz in u on the relevant region. Then*

$$\|R(u_t, h_s) - R(u_s, h_s)\| \leq L_R \|W\| a \sqrt{Cp^2 \tau_{\text{on}}}. \quad (\text{A8})$$

Proof. Inactivity implies patch MSE at most the applicable hysteresis threshold, which is at most τ_{on} . Dilation and forced activation only turn zeros into ones and preserve this implication for a final zero mask. Each consecutive patch difference is at most $\sqrt{Cp^2 \tau_{\text{on}}}$. Sum these differences, apply the embedding norm and then the readout bound. The cached readout at s used the already updated state h_s , which remains unchanged during the inactive interval. $\square$

The readout includes normalization, residual and input-dependent projections. This first-block, pixel-MSE statement does not directly apply to deeper-layer tokens or GMC relative-feature scores. Spatial active-subsequence computation has an additional context defect $\|S_A(u_A) - [S(u)]_A\|$, even on active tokens. For example, the spatial recurrence $s_i = \rho s_{i-1} + b_i$ with $b_1 = 1, b_2 = 0, s_0 = 0$ produces $s_2 = \rho$; omitting token 1 gives zero, even if the current inputs equal past inputs. Current GMC warps the comparison image, not the hidden-state or output caches.

Theorem A3 (Refresh is not state reset). *An all-active keyframe makes $G_t = F_t$ at the same incoming extended state; hence $d_t^{\text{net}} = 0$, but generally $E_t^{\text{net}} \leq L_t E_{t-1}^{\text{net}} \neq 0$. For the shared-input SSM of Theorem A1, assume a common $\rho < 1$, skip defects at most ϵ , and a refresh every K steps. If B_n is the error immediately after refresh n , then*

$$B_{n+1} \leq \rho^K B_n + \rho \epsilon \frac{1 - \rho^{K-1}}{1 - \rho}, \quad (\text{A9})$$

$$\limsup_{n \rightarrow \infty} B_n \leq \frac{\rho \epsilon (1 - \rho^{K-1})}{(1 - \rho)(1 - \rho^K)}. \quad (\text{A10})$$

Proof. All-active branches recompute block outputs but pass the carried temporal states to the recurrence. Thus only the same-state local defect vanishes. For the scalar norm recurrence, propagate $K - 1$ bounded defects and apply the final zero-defect refresh. Each injected term receives at least one factor ρ . Sum the resulting cycle recurrence as a geometric series. $\square$

At $j \leq K - 1$ steps after a refresh, the bound is $\rho^j B_n + \epsilon(1 - \rho^j)/(1 - \rho)$. Actual errors need not be monotone in K : changing K also changes the trajectories and defects. Keyframes bound cache age by $K - 1$, not dense-history error by zero. Resetting only one path to zero is a different intervention. True state synchronization would require the reference state or replay. Rolling refresh bounds per-patch age but is not an all-active whole-network refresh.

Appendix A.4. Fixed cost, refresh frequency and overhead

Proposition A3 (Conditional pipeline speedup). Assume an additive cost model with dense reference $C_d = F + V > 0$ and mean sparse cost $\bar{C}_s = F + \bar{a}_{\text{eff}}V + \bar{H}$, where $F, V, \bar{H} \geq 0$ and $\bar{C}_s > 0$. Define $f = F/C_d$ and $\omega = \bar{H}/C_d$. Then

$$S = \frac{C_d}{\bar{C}_s} = \frac{1}{f + (1 - f)\bar{a}_{\text{eff}} + \omega}, \quad S > 1 \iff \omega < (1 - f)(1 - \bar{a}_{\text{eff}}). \quad (\text{A11})$$

For $f > 0$, $S \leq 1/f$.

Proof. Divide by C_d and compare the positive denominator with one. Dropping its other nonnegative terms yields the upper bound. $\square$

This is an application of Amdahl-style fixed-cost reasoning [12], not a new universal speed law. Detector, gather/scatter, cache-copy and launch overheads must be counted without double counting. Keyframe and fallback events are a union: $a_{\text{eff},t} = 1$ on either event, and otherwise equals the normal activity. Padding and nonlinear kernel costs must be modeled separately if active fraction does not scale variable work linearly.

For T frames starting at frame index zero, the keyframe fraction is $\lceil T/K \rceil / T$, even on a static stream. At $T = 256, K = 30$ this is $9/256$, not exactly $1/30$. In a long stream without fallback and with constant nonkeyframe mean a , the effective activity tends to $a + (1 - a)/K$. If a verified cache-only coefficient $c = L_R \|W\| \sqrt{Cp^2\tau} > 0$ and tolerance E_c are available, (A8) gives the sufficient condition $K \leq 1 + \lfloor E_c/c \rfloor$. In the nondegenerate fixed-cost model, a target speedup $S_* > 1$ requires

$$D = 1/S_* - f - (1 - f)a - \omega > 0, \quad K \geq \left\lceil \frac{(1 - f)(1 - a)}{D} \right\rceil. \quad (\text{A12})$$

Nonoverlap means these sufficient error conditions cannot certify a feasible K ; it is not a proof that no accurate fast policy exists.

The existing K30 synchronized profile assigns about 69.4% of its time to image read/decode/preprocessing. Holding that component fixed and deleting all remaining profiled time gives an idealized ceiling of about $1.44\times$ relative to that same profile. This is neither a measured speedup nor a bound for a redesigned input path or GPU-resident pipeline. Main matched-subset latencies remain 7.496 ms sparse and 7.521 ms dense. Persistent temporal states are densely allocated:

$$bBN \sum_{\ell \in \text{temporal}} P_\ell S_\ell$$

bytes, independent of activity. For v11 this is 12 MiB in FP32; output caches add storage.

Appendix B. Reproduction Details and Evidence Boundaries

Appendix B.1. Checkpoint, training record and interventions

The archived effective configuration beside the seed-0 v11 checkpoint records 8000 final-stage steps, clip length 24, batch size 2, image size 256, learning rate 3×10^{-4} , 1000

warmup steps, gradient-norm clipping at 1 and EMA decay 0.999. The checkpoint resumes partially from v10. The recorded loss weights are multiscale gradient 0.5, normal 0.05, warp 2, spread 0.5, edge 2, and bin classification 0.2. Teacher and feature-distillation weights are zero in this stage only. Training uses random masks with maximum skip fraction 0.5; the image-change detector is not used to generate these training masks.

The archived training implementation uses AdamW (default weight decay 0.01), warmup followed by cosine learning-rate decay, and source-balanced weighted sampling over TUM, Bonn, VKITTI2, TartanAir2 and PointOdyssey. Clip-consistent augmentation comprises zoom/crop with scale 0.55–1.0, horizontal flip with probability 0.5, brightness/contrast factors 0.75–1.3, saturation 0.7–1.4, and gamma 0.8–1.25. The effective v11 sidecar records augmentation enabled and the dataset paths, but not all of these sampler/optimizer implementation details or the RNG trajectory. These are reproduction settings read from the archived implementation, not an independently reconstructed historical run. The sidecar’s recorded source commit is 0a3f5e06abbf448898a65c0a42c53534531912e5. The four-stage ancestry, rather than the earlier three-stage summary, must be used when describing training cost. Full-training seed variation remains unknown.

Only one reference operating configuration is selected: K30, thresholds 0.05/0.025, fallback 0.4, both caches on and GMC off. K5, dense carry, dense reset and output hold are frozen diagnostic controls, not post-test deployment recommendations. The development cache/readout interventions are available in the study-5–8 tables. Historical threshold/fallback sweeps remain in the older draft’s Table 9, not a newly validated frozen-protocol sensitivity panel. None of these should be mixed with final-test scores. A token-drop control zeroes the inactive temporal residual contribution (retaining the identity bypass and frozen hidden state); it is not equivalent to dropping input video frames. No universal accuracy gain over that control is claimed.

Appendix B.2. Temporal-score equations and masking

Let D_t be depth after the declared clip gauge and G_t ground truth. RAFT Small uses RGB mapped from $[0, 1]$ to $[-1, 1]$ at the common evaluation resolution and its final refinement output, with the default pretrained weights resolved by the frozen dependency record. Flow maps the current frame to the preceding frame. Let W_t warp that preceding frame with nearest-neighbor sampling, border padding and aligned grid corners. The common mask m_t is the product of the in-bounds indicator, current GT validity and warped previous GT validity. There is no forward-backward flow-consistency or semantic occlusion filter. With $N_m = \max(1, \sum_{t,i} m_{t,i})$,

$$\text{OPW} = \frac{1}{N_m} \sum_{t,i} m_{t,i} \frac{|(W_t D_{t-1})_i - D_{t,i}|}{\max(G_{t,i}, 10^{-6})}, \quad (\text{A13})$$

$$\text{TCE} = \frac{1}{N_m} \sum_{t,i} m_{t,i} \frac{|(W_t D_{t-1})_i - D_{t,i} - (W_t G_{t-1})_i + G_{t,i}|}{\max(G_{t,i}, 10^{-6})}. \quad (\text{A14})$$

Both are dimensionless. Raw temporal delta is the mean absolute consecutive depth difference over all pixels, without warping or a GT-validity mask, after the declared clip alignment; it has depth units and is not itself a ground-truth temporal-accuracy measure. Temporal scores are not reported after per-frame fitting, which would change the temporal signal. All code-level clamps and invalid-fit conventions remain those of the frozen scorer.

Appendix B.3. Memory, edge evidence and portability

For N independent streams, persistent storage is $W + NS$, before input/output buffers, temporary activations and allocator overhead. The native FP32 weights occupy about 15.968 MiB; K30 state plus caches occupies 13.501 MiB per stream, versus 12.000 MiB for dense carry without output caches. This is not peak VRAM, and inactive states are still densely allocated. Fewer parameters or updates therefore do not by themselves establish lower latency or multi-stream memory.

Historical Nano Developer Kit B01 logs cover user-confirmed 5 W and 10 W modes, but use a reference scan, fixed-ratio activity and repeated synthetic inputs. They are not the fused desktop path or the measured PNG-to-depth pipeline. The power source is module VDD_IN, not whole-system wall power. Missing historical weight/code linkage and independent power repetitions prevent those logs from establishing final-model real-video energy savings. Their separate audit is supplied without a quantitative edge claim in this paper.

The private package includes source, derived scores, manifests, configuration and native weights; original media and third-party baseline weights require separate authorized acquisition. Frozen paths may need a versioned relocation manifest on another machine. Full clean-machine training/inference reproduction and independent human verification have not been performed. The complete 40-item internal-review response is provided separately; partial and withdrawn items must not be read as experimentally resolved.

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